



Hybrid Expansion Cosmology in $f(T)$ Gravity: Late-Time Evolution and Observational Bounds

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Abstract

The cosmological expansion history of the exponential $f(T)$ gravity model proposed by Linder is determined from a modified Friedmann equation that generally does not admit a closed-form analytical solution for the Hubble parameter, requiring repeated numerical evaluations in cosmological analyses. Motivated by this limitation, we present an analytical reconstruction of the model by introducing a Lambert–W-based hybrid expansion ansatz for the Hubble parameter. The free parameters of the reconstruction are first determined through a least-squares optimization by minimizing the deviation from the numerical solution of the modified Friedmann equation, yielding a maximum relative difference of approximately 1.08% over the redshift range $0 \leq z \leq 4$. The reconstructed model is subsequently treated as an independent analytical cosmological model and constrained using recent late-time observations, including Cosmic Chronometer (CC), DESI DR2 Baryon Acoustic Oscillation (BAO), Pantheon+SH0ES, and Union3 Type Ia supernova datasets through Markov Chain Monte Carlo (MCMC) analyses. The resulting parameter constraints are mutually consistent across the individual datasets and become significantly tighter for the combined likelihood. Furthermore, various cosmological diagnostics demonstrate that the reconstructed model successfully describes the transition from an early decelerated universe to the present accelerated epoch while remaining in close agreement with the standard Λ CDM cosmology at late times. These results demonstrate that the proposed hybrid ansatz accurately reproduces the numerical exponential $f(T)$ expansion history while providing a computationally efficient analytical framework for cosmological parameter estimation and future observational studies.

Keywords

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Teleparallel gravity; FLRW metric; Hybrid scale factor; analytical reconstruction;
Late-time cosmic evolution

Introduction

Modern cosmology continues to confront profound challenges in providing a coherent explanation for the accelerated expansion of the late-time universe. Cosmological observations from various supernova-like type Ia Supernovae (SNe) Ia Riess, 1998, Riess, 2004, the Cosmic Microwave Background Radiation (CMBR) Komatsu (2009), and Large Scale Structure (LSS) Eisenstein (2005); Seljak (2005); Tegmark (2004) collectively indicate that the universe is presently experiencing an accelerated expansion during its late-time evolution Cai (2016); Hinshaw (2013); Perlmutter (1999); Riess (1998); Wang (2008). This phenomenon is generally attributed to an elusive component known as dark energy, which is characterized by a fluid with negative pressure. It is widely accepted that dark energy constitutes nearly two-thirds of the total energy of the universe. Recent observational evidence indicates that the combined contribution of dark energy and dark matter constitutes roughly 95 – 96% of the total energy budget of the universe Weinberg (1989). Among the numerous theoretical proposals put forward to describe dark energy, the cosmological constant (Λ) remains the most straightforward and widely supported candidate. The resulting Λ CDM model demonstrates remarkable agreement with a wide range of observational data. Although the cosmological constant provides a simple explanation for cosmic acceleration, it is accompanied by two significant theoretical issues, commonly referred to as the fine-tuning problem and the coincidence problem. Carroll (2001); Copeland et al. (2006a); Di Valentino et al. (2021); Padmanabhan (2003). These persistent issues have encouraged the investigation of alternative directions in cosmology, prompting the development of models that extend beyond the conventional framework.

To address these challenges, researchers have explored a variety of possible directions. One line of work introduces additional or non-standard forms of matter and energy to account for the observed acceleration. Another, more structural approach involves modifying Einstein's general relativity itself. The latter has drawn considerable attention, mainly because it offers the possibility that the accelerated expansion could arise from the gravitational sector, without the need to introduce a separate dark energy component. General relativity, despite its remarkable success in explaining gravitational phenomena over a wide range of scales, does face difficulties when applied to late-time cosmic acceleration. This has motivated the development of modified gravity theories that attempt to extend or go beyond the standard framework Clifton et al. (2012); Nojiri et al. (2017). In many cases, these theories are constructed by altering the curvature-based Einstein–Hilbert action, leading to modified field equations that can naturally accommodate an accelerating universe. Prominent examples include $f(R)$ gravity De Felice and Tsujikawa (2010); Nojiri and Odintsov (2011); Sotiriou and Faraoni (2010), $f(R, T)$ gravity Harko (2011); Myrzakulov (2012), $f(R, L_m)$ gravity Devi et al. (2024); F. S. N. Lobo (2010); S. A. Narawade and Mishra (2025) $f(G)$ gravity De Felice and Tsujikawa (2009), $f(R, G)$ gravity Nojiri and Odintsov (2005), $f(Q)$ gravity Heisenberg (2024); Koussour et al. (2022), $f(T, B)$ gravity Bahamonde et al. (2015); Bahamonde et al. (2023), $f(Q, T)$ gravity Xu et al. (2019) and $f(T)$ gravity Aldrovandi and Pereira (2012); Bahamonde et al. (2023); Bamba et al., 2012, Bamba et al., 2013; Bengochea and Ferraro (2009); Cardone et al. (2012). Each

of these approaches provides a distinct perspective on gravitational interactions and the overall evolution of the universe. The explanations put forward to account for the accelerated expansion of the universe can be divided into two principal categories. One possibility is to extend the matter sector by introducing additional dynamical fields, such as scalar fields (both canonical and phantom) or even vector fields, which can drive the acceleration Bassett et al. (2006); Copeland et al. (2006b). The other approach is to reconsider the gravitational framework itself. In this case, gravity can be formulated in different but equivalent geometric ways, based on curvature, torsion, or non-metricity Capozziello et al. (2011); Di Valentino et al. (2025). Within this latter category, gravity can be described through three equivalent geometrical frameworks, such as curvature, torsion, and non-metricity Beltran Jiménez et al. (2019).

Among these, torsion-based modifications offer a particularly promising direction due to their distinct geometric structure and the novel physical insights they provide. In particular, the torsion-based formulation has been extensively studied in recent years. In contrast to curvature-based gravity, where gravitational effects arise from the curvature of spacetime, the teleparallel formulation attributes gravitation to spacetime torsion. In this framework, gravity is described through the Teleparallel Equivalent of General Relativity (TEGR), which provides a formulation dynamically equivalent to General Relativity at the level of the field equations, while employing torsion instead of curvature as the fundamental geometrical quantity. This viewpoint opens new avenues for constructing modified theories of gravity that are dynamically distinct from their curvature-based counterparts. In particular, the extension to $f(T)$ gravity has emerged as one of the most extensively studied modifications, owing to its ability to modify gravitational dynamics while preserving second-order field equations Bahamonde et al. (2023). This feature makes it mathematically simpler and computationally more tractable compared to curvature-based extensions such as $f(R)$ gravity. Moreover, $f(T)$ gravity provides a compelling framework for explaining the observed late-time acceleration of the universe without introducing an explicit dark energy component. Recent studies have investigated its cosmological implications and observational consistency. Several recent studies have explored the cosmological implications of extended teleparallel gravity models Kofinas and Saridakis (2014); Lohakare et al. (2023). In particular, Lohakare et al. (2023) investigated an extended $f(T, T_C)$ gravity framework constrained by observational datasets, demonstrating that the model successfully reproduces the cosmic expansion history. Their analysis predicts a smooth transition from an early decelerating phase to a late-time accelerating universe with a quintessence-like behavior, while also satisfying the energy conditions at the background level. SK Tripathy et al. (2024) develop a scalar-field extension of teleparallel gravity to model bounce cosmology and test its physical consistency. In Kumar et al. (2023a), the authors investigate the viability of the power-law ($f(T)$) gravity model using a combination of recent cosmological observations, including the Planck 2018 Cosmic Microwave Background (CMB), Pantheon Type Ia Supernovae, Baryon Acoustic Oscillations (BAO), and Big Bang Nucleosynthesis (BBN) datasets. To perform a comprehensive analysis, they consider both the background expansion history of the universe and the evolution of linear cosmological perturbations. This approach enables a more stringent comparison between the theoretical predictions of the model and current observational data, leading to tighter constraints on the model parameters Nunes (2018). The study explores the impact of modified $f(T)$ gravity on the evolution of cosmic structures and the CMB through scalar perturbations. The analysis shows that

the model agrees well with current cosmological observations while offering a potential resolution to the Hubble tension. Within the torsion-based formulation, the Teleparallel Equivalent of General Relativity (TEGR) provides an alternative description of gravitation. In this framework, torsion replaces the curvature of the fundamental quantity governing gravitational interactions, and the tetrad fields serve as the fundamental dynamical variables instead of the metric tensor. The tetrad basis $e_A(x^\mu)$, where $A = 0, 1, 2, 3$ labels the tangent-space coordinates and $\mu = 0, 1, 2, 3$ denotes spacetime indices, forms a set of orthonormal vectors. Each Vector e_A can be decomposed into a coordinate basis and produces the components $e^\mu_A \partial_\mu$. These tetrad fields form an orthonormal set of four linearly independent basis vectors defined in the tangent space at each point in spacetime. Hence, the orthogonality condition becomes

$$g_{\mu\nu} = \eta_{AB} e^\mu_A e^\nu_B.$$

Where $g_{\mu\nu}$ is the metric tensor and $\eta_{AB} = \text{diag}(1, -1, -1, -1)$ is the Minkowski metric. The co-frame $\{e^\mu_A\}$ represents the inverse of the basis, enabling us to obtain the relation as

$$e^\mu_A e^\nu_B = \delta^\mu_B$$

and

$$e^\mu_A e^\nu_B = \delta^\nu_A$$

An important extension of TEGR is provided as $f(T)$ gravity, which has been proposed as a viable framework for explaining late-time cosmological phenomena in the universe. In contrast to GR, where gravitational interactions are governed by the curvature scalar R , the teleparallel formulation is based on the torsion scalar T . Within this approach, the $f(T)$ gravity is constructed by promoting the torsion scalar to a generalized function, thereby modifying the standard teleparallel Lagrangian density. As a result, the theory extends the TEGR action from a linear dependence on T to a more general form involving $T + f(T)$, allowing for a richer corresponding action can be written as:

$$S = \frac{1}{16\pi G} \int d^4x e [T + f(T) + \mathcal{L}_m]$$

In the above action, $\mathcal{L}_m$ the total matter Lagrangian, and $e = \det[e^\mu_A] = \sqrt{-g}$ represent the determinant of the tetrad fields. Throughout this work, we adopt natural units such that $8\pi G = \kappa^2 = 1$. Within the teleparallel framework, the gravitational action is constructed from the torsion scalar T , in contradiction to GR, which is formulated using the torsionless Levi-Civita connection and the curvature scalar R . Teleparallel Gravity, on the other hand, employs the curvature-free Witz-connection to describe gravitational interaction. As a result, gravity is attributed to space-time torsion rather than curvature. In this formulation, the fundamental

connection is defined as

$$\hat{\Gamma}_{\mu\nu}^{\lambda} \equiv e^{\lambda}_A \partial_{\mu} e^A_{\nu}$$

Within the Weitzenböck connection, the curvature tensor identically vanishes, implying that spacetime is described by zero curvature while torsion remains non-zero. Based on this framework, the torsion tensor can be expressed as follows

$$T^{\lambda}_{\mu\nu} \equiv \hat{\Gamma}^{\lambda}_{\nu\mu} - \hat{\Gamma}^{\lambda}_{\mu\nu} = e^{\lambda}_A (\partial_{\mu} e^A_{\nu} - \partial_{\nu} e^A_{\mu}).$$

Where $\hat{\Gamma}^{\lambda}_{\mu\nu}$ is the connection coefficient of the Wittenberg connection.

Further, the contortion tensor can be defined as

$$K^{\mu\nu}_{\rho} \equiv \frac{1}{2} (T^{\nu\mu}_{\rho} + T^{\mu\nu}_{\rho} - T^{\mu\nu}_{\rho})$$

From the above, we can define the superpotential tensor

$$S_{\rho}^{\mu\nu} \equiv \frac{1}{2} (K^{\mu\nu}_{\rho} + \delta^{\mu}_{\rho} T^{\alpha\nu}_{\alpha} - \delta^{\nu}_{\rho} T^{\alpha\mu}_{\alpha}),$$

Using the above, we can write the teleparallel Lagrangian density, which is the torsion scalar. The torsion scalar is defined through an appropriate contraction of the torsion tensor. We can write it as

$$T \equiv S_{\rho}^{\mu\nu} T^{\rho}_{\mu\nu}$$

A considerable body of work has been devoted to exploring cosmological solutions within the framework of $f(T)$ gravity, particularly in relation to the dynamical evolution of the universe Paliathanasis et al. (2016). In addition, investigations into the thermodynamic properties of these models have been carried out Salako et al. (2013), while classical energy conditions have been examined to assess their physical viability. A key feature of $f(T)$ gravity is that the resulting field equations are of second order, unlike those in $f(R)$ gravity, which generally involve fourth-order differential equations. This feature makes the theory comparatively simpler and more tractable for cosmological applications. The $f(T)$ framework has been widely employed in both astrophysical and cosmological studies, offering an alternative route to explain the late-time accelerated expansion of the universe without explicitly introducing a dark energy component. The gravitational field equations are derived by performing a variation of the action (2) with respect to the vierbein, leading to

$$e^{-1} \partial_{\mu} (e e^{\rho}_A S^{\mu\nu}_{\rho}) [1 + f_T] + e^{\rho}_A S^{\mu\nu}_{\rho} \partial_{\mu} (T) f_{TT} - e^{\lambda}_A T^{\rho}_{\mu\lambda} S^{\nu\mu}_{\rho} [1 + f_T] + \frac{1}{4} e^{\nu}_A [T + f(T)] = 4\pi G e^{\rho}_A \mathcal{T}^{\nu}_{\rho},$$

We consider f to be a function of the torsion scalar, written as $f(T)$, where the quantities $f_T = \frac{df}{dT}$ and $f_{TT} = \frac{d^2f}{dT^2}$ represent its first and second derivatives with

respect to T , respectively. In addition, $\mathcal{T}^\rho_\nu$ represents the total matter energy-momentum tensor constructed by the Lagrangian matter field.

The structure of this paper is outlined as follows. In Section 2, we introduce the hybrid scale factor ansatz within the framework of teleparallel $f(T)$ gravity and derive the corresponding Hubble parameter in terms of the Lambert- W function. Section 3 presents the exponential $f(T)$ gravity model proposed by Linder, along with the power-law and exponential $f(T)$ parameterizations considered for comparison. In Section 4, we validate the proposed analytical Lambert- W hybrid reconstruction by comparing its expansion history with the numerical solution of the original exponential $f(T)$ model and with other analytical parameterizations. Section 5 is devoted to the observational constraints on the reconstructed model using Cosmic Chronometer (CC), DESI DR2 Baryon Acoustic Oscillation (BAO), Pantheon+SH0ES, and Union3 Type Ia supernova datasets, together with their combined analysis. Section 6 investigates the physical and dynamical properties of the reconstructed model through the deceleration parameter, effective equation-of-state parameter, statefinder diagnostics, $Om(z)$ diagnostic, and the energy conditions. Finally, Section 7 presents the conclusion and discussion, summarizing the main results and outlining possible directions for future investigations.

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Hybrid Scale Factor Ansatz

For the analysis of the cosmological framework, we adopt the spatially flat Friedmann-Lemaître-Robertson-Walker (FLRW) metric expressed in Cartesian coordinates.

$$ds^2 = dt^2 - a(t)^2 (dx^2 + dy^2 + dz^2),$$

Here, $a(t)$ is the scaling factor, and the tetrad is chosen as

$$e^A_\mu = \text{diag}(1, a(t), a(t), a(t)).$$

For the spatially flat FLRW geometry, the torsion scalar is given by

$$T = -6H^2.$$

Varying the action with respect to the tetrad yields the modified Friedmann equations

$$3H^2 = \kappa\rho + \frac{1}{2}(2Tf_T - f)$$

$$2\dot{H} = -\frac{\kappa(\rho + p)}{1 + f_T + 2Tf_{TT}}$$

The Hubble parameter is ...

Cosmological $f(T)$ Models

Among the various modified teleparallel gravity models, exponential $f(T)$ gravity has attracted considerable attention for introducing smooth deviations from TEGR while remaining compatible with cosmological observations and local gravity constraints Cai (2016); Linder (2010), and has been successfully tested against Type Ia supernovae, baryon acoustic oscillations, cosmic chronometers, and CMB data Basilakos et al. (2013); Kumar et al. (2023b); Nunes (2018); Nunes et al. (2016). Motivated by ...

Validation of the Proposed Analytical Reconstruction

The principal objective of the present work is to develop and validate an analytical reconstruction of the exponential $f(T)$ gravity model proposed by LinderLinder (2010) through the introduction of a hybrid scale factor ansatz. Unlike the original exponential $f(T)$ model, whose cosmological expansion history is determined from a modified Friedmann equation that generally does not admit a closed-form analytical solution for $H(z)$, the proposed reconstruction provides an explicit analytical ...

Observational Constraints

Following the successful validation of the proposed hybrid Lambert-W reconstruction against the numerical exponential $f(T)$ gravity model, we proceed to investigate its compatibility with current cosmological observations. While the previous section established the capability of the analytical ansatz to reproduce the numerical expansion history with high accuracy, the present analysis aims to determine whether the reconstructed model is also favoured by observational data. To this end, the free ...

Deceleration Parameter

The deceleration parameter is one of the most fundamental cosmological quantities used to characterize the expansion history of the universe. It provides a direct measure of whether the cosmic expansion is accelerating or decelerating and is

defined as

$$q(z) = -1 + \frac{(1+z)}{H(z)} \frac{dH(z)}{dz}.$$

For the proposed hybrid Lambert-W reconstruction, the above expression admits the following analytical form

$$q(z) = \frac{1}{\beta[1+W(x)]^2} - 1,$$

where

$$x = \frac{\lambda}{\beta}(1+z)^{-1/\beta},$$

and $W(x)$ denotes the principal branch of the Lambert-W function. An important ...

Conclusion and Discussion

In this work, we have presented an analytical reconstruction of the exponential $f(T)$ gravity model by introducing a hybrid expansion ansatz capable of accurately reproducing the late-time cosmological evolution of the original numerical solution. The principal motivation of this reconstruction is to provide an analytical description of the cosmological expansion that eliminates the need for repeated numerical integration of the modified Friedmann equations while preserving the essential ...

Declaration

Conflicts of Interest: The authors have no conflict of interest. ...

Uncited floats

Fig. 6, Fig. 8 ...

CRedit authorship contribution statement

Rajalaxmi Jena: Writing – original draft. **Vishal M C:** Writing – original draft.

Sankarsan Tarai: Writing – review & editing, Writing – original draft, Investigation, Conceptualization. ...

Declaration of competing interest

The authors declare no conflict of interest ...

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